Newman-Penrose Formalism Calculations for

metric:

Name: Stephani et al., Equation (17.26)

Alias: Stephanietal_17_26

Reference(s): [**?**]

Line element:

$$\begin{aligned} ds^2 = & -r^2 B\left(y\right) e^{\left(\Lambda F(r,x)-\Lambda G(r,y)\right)} \mathrm{d}t \otimes \mathrm{d}t + r^2 A\left(x\right) e^{\left(-\Lambda F(r,x)+\Lambda G(r,y)\right)} \mathrm{d}z \otimes \mathrm{d}z \\ & + \frac{r^2 e^{\left(-\Lambda^2 r^4 A(x) B(y)+\Lambda F(r,x)+\Lambda G(r,y)\right)}}{A\left(x\right)} \mathrm{d}x \otimes \mathrm{d}x + \frac{r^2 e^{\left(-\Lambda^2 r^4 A(x) B(y)+\Lambda F(r,x)+\Lambda G(r,y)\right)}}{B\left(y\right)} \mathrm{d}y \otimes \mathrm{d}y \end{aligned} \tag{1}$$

Coordinates: (t, z, x, y)

Null tetrad components:

Covariant components:

$$l = \left(\sqrt{\frac{1}{2}} r \sqrt{B\left(y\right)} e^{\left(\frac{1}{2}\Lambda(F(r,x)-G(r,y))\right)}\right) dt + \left(-\sqrt{\frac{1}{2}} r \sqrt{A\left(x\right)} e^{\left(-\frac{1}{2}\Lambda(F(r,x)-G(r,y))\right)}\right) dz + (0)\, dx + (0)\, dy \tag{2}$$

$$n = \left(\sqrt{\frac{1}{2}} r \sqrt{B\left(y\right)} e^{\left(\frac{1}{2}\Lambda(F(r,x)-G(r,y))\right)}\right) dt + \left(\sqrt{\frac{1}{2}} r \sqrt{A\left(x\right)} e^{\left(-\frac{1}{2}\Lambda(F(r,x)-G(r,y))\right)}\right) dz + (0)\, dx + (0)\, dy \tag{3}$$

$$m = (0)\, dt + (0)\, dz + \left(\frac{\sqrt{\frac{1}{2}} r e^{\left(-\frac{1}{2}\Lambda^2 r^4 A(x) B(y)+\frac{1}{2}\Lambda(F(r,x)+G(r,y))\right)}}{\sqrt{A\left(x\right)}}\right) dx + \left(\frac{i\sqrt{\frac{1}{2}} r e^{\left(-\frac{1}{2}\Lambda^2 r^4 A(x) B(y)+\frac{1}{2}\Lambda(F(r,x)+G(r,y))\right)}}{\sqrt{B\left(y\right)}}\right) dy \tag{4}$$

$$\overline{m} = (0)\,dt + (0)\,dz + \left(\frac{\sqrt{\frac{1}{2}}re^{\left(-\frac{1}{2}\,\Lambda^2 r^4 A(x)B(y)+\frac{1}{2}\,\Lambda(F(r,x)+G(r,y))\right)}}{\sqrt{A\,(x)}}\right)dx + \left(-\frac{i\,\sqrt{\frac{1}{2}}re^{\left(-\frac{1}{2}\,\Lambda^2 r^4 A(x)B(y)+\frac{1}{2}\,\Lambda(F(r,x)+G(r,y))\right)}}{\sqrt{B\,(y)}}\right)dy \tag{5}$$

Contravariant components:

$$l = \left(-\frac{\sqrt{2}e^{\left(-\frac{1}{2}\,\Lambda F(r,x)+\frac{1}{2}\,\Lambda G(r,y)\right)}}{2\,r\sqrt{B\,(y)}}\right)dt + \left(-\frac{\sqrt{2}e^{\left(\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}}{2\,r\sqrt{A\,(x)}}\right)dz + (0)\,dx + (0)\,dy \tag{6}$$

$$n = \left(-\frac{\sqrt{2}e^{\left(-\frac{1}{2}\,\Lambda F(r,x)+\frac{1}{2}\,\Lambda G(r,y)\right)}}{2\,r\sqrt{B\,(y)}}\right)dt + \left(\frac{\sqrt{2}e^{\left(\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}}{2\,r\sqrt{A\,(x)}}\right)dz + (0)\,dx + (0)\,dy \tag{7}$$

$$m = (0)\,dt + (0)\,dz + \left(\frac{\sqrt{2}\sqrt{A\,(x)}e^{\left(\frac{1}{2}\,\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}}{2\,r}\right)dx + \left(\frac{i\,\sqrt{2}\sqrt{B\,(y)}e^{\left(\frac{1}{2}\,\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}}{2\,r}\right)dy \tag{8}$$

$$\overline{m} = (0)\,dt + (0)\,dz + \left(\frac{\sqrt{2}\sqrt{A\,(x)}e^{\left(\frac{1}{2}\,\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}}{2\,r}\right)dx + \left(-\frac{i\,\sqrt{2}\sqrt{B\,(y)}e^{\left(\frac{1}{2}\,\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}}{2\,r}\right)dy \tag{9}$$

Directional derivatives:

$$D = -\frac{\sqrt{2}e^{\left(-\frac{1}{2}\,\Lambda F(r,x)+\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial t}f\left(t,z,x,y\right)}{2\,r\sqrt{B\left(y\right)}} - \frac{\sqrt{2}e^{\left(\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial z}f\left(t,z,x,y\right)}{2\,r\sqrt{A\left(x\right)}} \tag{10}$$

$$\Delta = -\frac{\sqrt{2}e^{\left(-\frac{1}{2}\,\Lambda F(r,x)+\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial t}f\left(t,z,x,y\right)}{2\,r\sqrt{B\left(y\right)}} + \frac{\sqrt{2}e^{\left(\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial z}f\left(t,z,x,y\right)}{2\,r\sqrt{A\left(x\right)}} \tag{11}$$

$$\delta = \frac{\sqrt{2}\sqrt{A\left(x\right)}e^{\left(\frac{1}{2}\,\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial x}f\left(t,z,x,y\right)}{2\,r} + \frac{i\,\sqrt{2}\sqrt{B\left(y\right)}e^{\left(\frac{1}{2}\,\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial y}f\left(t,z,x,y\right)}{2\,r} \tag{12}$$

$$\overline{\delta} = \frac{\sqrt{2}\sqrt{A\left(x\right)}e^{\left(\frac{1}{2}\,\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial x}f\left(t,z,x,y\right)}{2\,r} - \frac{i\,\sqrt{2}\sqrt{B\left(y\right)}e^{\left(\frac{1}{2}\,\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial y}f\left(t,z,x,y\right)}{2\,r} \tag{13}$$

Spin coefficients:

$$\rho = 0 \tag{14}$$

$$\sigma = 0 \tag{15}$$

$$\begin{aligned}\kappa = &-\frac{\sqrt{2}\Lambda\sqrt{A\left(x\right)}e^{\left(\frac{1}{2}\,\Lambda^{2}r^{4}A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F\left(r,x\right)}{4\,r}+\frac{i\,\sqrt{2}\Lambda\sqrt{B\left(y\right)}e^{\left(\frac{1}{2}\,\Lambda^{2}r^{4}A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G\left(r,y\right)}{4\,r}\\ &+\frac{\sqrt{2}e^{\left(\frac{1}{2}\,\Lambda^{2}r^{4}A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A\left(x\right)}{8\,r\sqrt{A\left(x\right)}}-\frac{i\,\sqrt{2}e^{\left(\frac{1}{2}\,\Lambda^{2}r^{4}A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B\left(y\right)}{8\,r\sqrt{B\left(y\right)}}\end{aligned} \tag{16}$$

$$\mu = 0 \tag{17}$$

$$\lambda = 0 \tag{18}$$

$$\begin{aligned}\nu = &\frac{\sqrt{2}\Lambda\sqrt{A\left(x\right)}e^{\left(\frac{1}{2}\,\Lambda^{2}r^{4}A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F\left(r,x\right)}{4\,r}+\frac{i\,\sqrt{2}\Lambda\sqrt{B\left(y\right)}e^{\left(\frac{1}{2}\,\Lambda^{2}r^{4}A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G\left(r,y\right)}{4\,r}\\ &-\frac{\sqrt{2}e^{\left(\frac{1}{2}\,\Lambda^{2}r^{4}A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A\left(x\right)}{8\,r\sqrt{A\left(x\right)}}-\frac{i\,\sqrt{2}e^{\left(\frac{1}{2}\,\Lambda^{2}r^{4}A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B\left(y\right)}{8\,r\sqrt{B\left(y\right)}}\end{aligned} \tag{19}$$

$$\begin{aligned}\alpha = &\frac{1}{8}\,\sqrt{2}\Lambda^{2}r^{3}\sqrt{A\left(x\right)}B\left(y\right)e^{\left(\frac{1}{2}\,\Lambda^{2}r^{4}A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A\left(x\right)-\frac{1}{8}i\,\sqrt{2}\Lambda^{2}r^{3}A\left(x\right)\sqrt{B\left(y\right)}e^{\left(\frac{1}{2}\,\Lambda^{2}r^{4}A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B\left(y\right)\\ &-\frac{\sqrt{2}\Lambda\sqrt{A\left(x\right)}e^{\left(\frac{1}{2}\,\Lambda^{2}r^{4}A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F\left(r,x\right)}{8\,r}+\frac{i\,\sqrt{2}\Lambda\sqrt{B\left(y\right)}e^{\left(\frac{1}{2}\,\Lambda^{2}r^{4}A(x)B(y)-\frac{1}{2}\,\Lambda F(r,x)-\frac{1}{2}\,\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G\left(r,y\right)}{8\,r}\end{aligned} \tag{20}$$

$$\beta = -\frac{1}{8}\sqrt{2}\Lambda^2 r^3 \sqrt{A(x)} B(y)\, e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) - \frac{1}{8} i \sqrt{2}\Lambda^2 r^3 A(x) \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) + \frac{\sqrt{2}\Lambda \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{8\,r} + \frac{i\sqrt{2}\Lambda \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{8\,r} \tag{21}$$

$$\tau = -\frac{\sqrt{2} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}} \tag{22}$$

$$\pi = \frac{\sqrt{2} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}} \tag{23}$$

$$\epsilon = 0 \tag{24}$$

$$\gamma = 0 \tag{25}$$

Relations between spin coefficients:

Ricci tensor components:

$$\Phi_{00} = \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) \frac{\partial}{\partial x} F(r,x)}{4\,r^2} + \frac{\Lambda A(x)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F(r,x)}{4\,r^2} - \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \frac{\partial}{\partial y} G(r,y)}{4\,r^2} - \frac{\Lambda B(y)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G(r,y)}{4\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A(x)}{8\,r^2} + \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B(y)}{8\,r^2} \tag{26}$$

$$\Phi_{01} = 0 \tag{27}$$

$$\Phi_{02} = -\frac{1}{8}\Lambda^2 r^2 B(y)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)^2 - \frac{1}{4} i\,\Lambda^2 r^2 \sqrt{A(x)}\sqrt{B(y)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) \frac{\partial}{\partial y} B(y) + \frac{1}{8}\Lambda^2 r^2 A(x)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)^2 - \frac{\Lambda^2 A(x)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)^2}{8\,r^2} + \frac{i\,\Lambda^2 \sqrt{A(x)}\sqrt{B(y)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x) \frac{\partial}{\partial y} G(r,y)}{4\,r^2} + \frac{\Lambda^2 B(y)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)^2}{8\,r^2} + \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) \frac{\partial}{\partial x} F(r,x)}{4\,r^2} - \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \frac{\partial}{\partial y} G(r,y)}{4\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A(x)}{8\,r^2} + \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B(y)}{8\,r^2} \tag{28}$$

$$\Phi_{10} = 0 \tag{29}$$

$$
\begin{aligned}
\Phi_{11} = {} & \frac{1}{16}\,\Lambda^2 r^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 + \frac{1}{8}\,\Lambda^2 r^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right) \\
& + \frac{1}{16}\,\Lambda^2 r^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 + \frac{1}{8}\,\Lambda^2 r^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right) \\
& - \frac{\Lambda^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{16\,r^2} - \frac{\Lambda^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{16\,r^2} \\
& - \frac{\Lambda A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{8\,r^2} - \frac{\Lambda B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{8\,r^2}
\end{aligned}
\tag{30}
$$

$$
\Phi_{12} = 0 \tag{31}
$$

$$
\begin{aligned}
\Phi_{20} = {} & -\frac{1}{8}\,\Lambda^2 r^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 + \frac{1}{4} i\,\Lambda^2 r^2 \sqrt{A\left(x\right)}\sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} B\left(y\right) \\
& + \frac{1}{8}\,\Lambda^2 r^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 - \frac{\Lambda^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{8\,r^2} \\
& - \frac{i\,\Lambda^2 \sqrt{A\left(x\right)}\sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} + \frac{\Lambda^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{8\,r^2} \\
& + \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} - \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} \\
& - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} + \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2}
\end{aligned}
\tag{32}
$$

$$
\Phi_{21} = 0 \tag{33}
$$

$$\Phi_{22} = \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) \frac{\partial}{\partial x} F(r,x)}{4\,r^2} + \frac{\Lambda A(x)\, e^{\left(\Lambda^2 r^4 A(x)B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F(r,x)}{4\,r^2}$$
$$- \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \frac{\partial}{\partial y} G(r,y)}{4\,r^2} - \frac{\Lambda B(y)\, e^{\left(\Lambda^2 r^4 A(x)B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G(r,y)}{4\,r^2}$$
$$- \frac{e^{\left(\Lambda^2 r^4 A(x)B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A(x)}{8\,r^2} + \frac{e^{\left(\Lambda^2 r^4 A(x)B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B(y)}{8\,r^2} \tag{34}$$

Weyl tensor components:

$$
\begin{aligned}
\Psi_0 = {} & \frac{1}{4}\,\Lambda^3 r^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right) + \frac{1}{4} i\,\Lambda^3 r^2 A\left(x\right)^{\frac{3}{2}} \sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial x} F\left(r,x\right) \\
& - \frac{1}{4} i\,\Lambda^3 r^2 \sqrt{A\left(x\right)} B\left(y\right)^{\frac{3}{2}} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} G\left(r,y\right) + \frac{1}{4}\,\Lambda^3 r^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right) \\
& - \frac{1}{8}\,\Lambda^2 r^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 - \frac{1}{8}\,\Lambda^2 r^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 \\
& - \frac{\Lambda^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{4\,r^2} - \frac{\Lambda^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{4\,r^2} \\
& + \frac{3\,\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{8\,r^2} + \frac{\Lambda A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} \\
& + \frac{3\,\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{8\,r^2} + \frac{\Lambda B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} \\
& - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2}
\end{aligned}
\tag{35}
$$

$$
\Psi_1 = 0 \tag{36}
$$

$$
\begin{aligned}
\Psi_2 = & -\frac{1}{24}\,\Lambda^2 r^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 - \frac{1}{12}\,\Lambda^2 r^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right) \\
& - \frac{1}{24}\,\Lambda^2 r^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 - \frac{1}{12}\,\Lambda^2 r^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right) \\
& - \frac{\Lambda^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{12\,r^2} - \frac{\Lambda^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{12\,r^2} \\
& + \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{8\,r^2} + \frac{\Lambda A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{12\,r^2} \\
& + \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{8\,r^2} + \frac{\Lambda B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{12\,r^2} \\
& - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{24\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{24\,r^2}
\end{aligned}
\tag{37}
$$

$$
\Psi_3 = 0 \tag{38}
$$

$$
\begin{aligned}
\Psi_4 = {} & \frac{1}{4}\,\Lambda^3 r^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right) - \frac{1}{4} i\,\Lambda^3 r^2 A\left(x\right)^{\frac{3}{2}} \sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial x} F\left(r,x\right) \\
& + \frac{1}{4} i\,\Lambda^3 r^2 \sqrt{A\left(x\right)} B\left(y\right)^{\frac{3}{2}} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} G\left(r,y\right) + \frac{1}{4}\,\Lambda^3 r^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right) \\
& - \frac{1}{8}\,\Lambda^2 r^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 - \frac{1}{8}\,\Lambda^2 r^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 \\
& - \frac{\Lambda^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{4\,r^2} - \frac{\Lambda^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{4\,r^2} \\
& + \frac{3\,\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{8\,r^2} + \frac{\Lambda A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} \\
& + \frac{3\,\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{8\,r^2} + \frac{\Lambda B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} \\
& - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2}
\end{aligned}
\tag{39}
$$

Newman-Penrose equations:

$$NP1 = 0 \tag{40}$$

$$NP2 = 0 \tag{41}$$

$$NP3 = 0 \tag{42}$$

$$NP4 = 0 \tag{43}$$

$$NP5 = 0 \tag{44}$$

$$NP6 = 0 \tag{45}$$

$$NP7 = 0 \tag{46}$$

$$NP8 = 0 \tag{47}$$

$$NP9 = 0 \tag{48}$$

$$NP10 = 0 \tag{49}$$

$$NP11 = 0 \tag{50}$$

$$NP12 = 0 \tag{51}$$

$$NP13 = 0 \tag{52}$$

$$NP14 = 0 \tag{53}$$

$$NP15 = 0 \tag{54}$$

$$NP16 = 0 \tag{55}$$

$$NP17 = 0 \tag{56}$$

$$NP18 = 0 \tag{57}$$

Bianchi identities:

$$BI1 = 0 \tag{58}$$

$$BI2 = 0 \tag{59}$$

$$BI3 = 0 \tag{60}$$

$$BI4 = 0 \tag{61}$$

$$BI5 = 0 \tag{62}$$

$$BI6 = 0 \tag{63}$$

$$BI7 = 0 \tag{64}$$

$$BI8 = 0 \tag{65}$$

$$BI9 = 0 \tag{66}$$

$$BI10 = 0 \tag{67}$$

$$BI11 = 0 \tag{68}$$

Petrov classification:
Petrov class: I

A Type transformations:

Spin coefficients:

$$\rho' = -\frac{\sqrt{2}\Lambda z\sqrt{A(x)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)}{4r} + \frac{i\sqrt{2}\Lambda z\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)}{4r} + \frac{\sqrt{2}ze^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8r\sqrt{A(x)}} - \frac{i\sqrt{2}ze^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8r\sqrt{B(y)}} \tag{69}$$

$$\sigma' = -\frac{\sqrt{2}\Lambda z\sqrt{A(x)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)}{4r} + \frac{i\sqrt{2}\Lambda z\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)}{4r} + \frac{\sqrt{2}ze^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8r\sqrt{A(x)}} - \frac{i\sqrt{2}ze^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8r\sqrt{B(y)}} \tag{70}$$

$$\kappa' = \kappa \tag{71}$$

$$\mu' = -\frac{\sqrt{2}\Lambda z^3\sqrt{A(x)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)}{4r} + \frac{i\sqrt{2}\Lambda z^3\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)}{4r} + \frac{\sqrt{2}z^3e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8r\sqrt{A(x)}} - \frac{i\sqrt{2}z^3e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8r\sqrt{B(y)}} + \frac{\sqrt{2}ze^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8r\sqrt{A(x)}} - \frac{i\sqrt{2}ze^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8r\sqrt{B(y)}} \tag{72}$$

$$\lambda' = -\frac{\sqrt{2}\Lambda z\sqrt{A(x)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)}{4r} + \frac{i\sqrt{2}\Lambda z\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)}{4r} + \frac{\sqrt{2}ze^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8r\sqrt{A(x)}} - \frac{i\sqrt{2}ze^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8r\sqrt{B(y)}} \tag{73}$$

$$
\begin{aligned}
\nu' = & -\frac{\sqrt{2}\Lambda z^3 \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{4\,r} + \frac{i\sqrt{2}\Lambda z^3 \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{4\,r} \\
& + \frac{\sqrt{2} z^3 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2} z^3 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}} \\
& + \frac{\sqrt{2}\Lambda \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{4\,r} + \frac{i\sqrt{2}\Lambda \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{4\,r} \\
& - \frac{\sqrt{2} z e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{3i\sqrt{2} z e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}} \\
& - \frac{\sqrt{2} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}}
\end{aligned}
\tag{74}
$$

$$
\begin{aligned}
\alpha' = & \frac{1}{8}\sqrt{2}\Lambda^2 r^3 \sqrt{A(x)} B(y)\, e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) - \frac{1}{8} i\sqrt{2}\Lambda^2 r^3 A(x) \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \\
& - \frac{\sqrt{2}\Lambda z^2 \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{4\,r} + \frac{i\sqrt{2}\Lambda z^2 \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{4\,r} \\
& + \frac{\sqrt{2} z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2} z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}} \\
& - \frac{\sqrt{2}\Lambda \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{8\,r} + \frac{i\sqrt{2}\Lambda \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{8\,r}
\end{aligned}
\tag{75}
$$

$$
\begin{aligned}
\beta' = & -\frac{1}{8}\sqrt{2}\Lambda^2 r^3 \sqrt{A(x)} B(y)\, e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) - \frac{1}{8} i\sqrt{2}\Lambda^2 r^3 A(x) \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \\
& - \frac{\sqrt{2}\Lambda z^2 \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{4\,r} + \frac{i\sqrt{2}\Lambda z^2 \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{4\,r} \\
& + \frac{\sqrt{2} z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2} z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}} \\
& + \frac{\sqrt{2}\Lambda \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{8\,r} + \frac{i\sqrt{2}\Lambda \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x) B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{8\,r}
\end{aligned}
\tag{76}
$$

$$\tau' = -\frac{\sqrt{2}\Lambda z^2\sqrt{A(x)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)}{4\,r} + \frac{i\sqrt{2}\Lambda z^2\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)}{4\,r}$$
$$+\frac{\sqrt{2}z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2}z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8\,r\sqrt{B(y)}}$$
$$-\frac{\sqrt{2}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8\,r\sqrt{B(y)}} \tag{77}$$

$$\pi' = -\frac{\sqrt{2}\Lambda z^2\sqrt{A(x)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)}{4\,r} + \frac{i\sqrt{2}\Lambda z^2\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)}{4\,r}$$
$$+\frac{\sqrt{2}z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2}z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8\,r\sqrt{B(y)}}$$
$$+\frac{\sqrt{2}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8\,r\sqrt{B(y)}} \tag{78}$$

$$\epsilon' = -\frac{\sqrt{2}\Lambda z\sqrt{A(x)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)}{4\,r} + \frac{i\sqrt{2}\Lambda z\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)}{4\,r}$$
$$+\frac{\sqrt{2}z e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2}z e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8\,r\sqrt{B(y)}} \tag{79}$$

$$
\begin{aligned}
\gamma' = & -\frac{1}{8}\sqrt{2}\Lambda^2 r^3 z\sqrt{A(x)}B(y)\,e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x) - \frac{1}{8}i\sqrt{2}\Lambda^2 r^3 z A(x)\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y) \\
& - \frac{\sqrt{2}\Lambda z^3\sqrt{A(x)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)}{4\,r} + \frac{i\sqrt{2}\Lambda z^3\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)}{4\,r} \\
& + \frac{\sqrt{2}z^3 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2}z^3 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8\,r\sqrt{B(y)}} \\
& + \frac{\sqrt{2}\Lambda z\sqrt{A(x)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)}{8\,r} + \frac{i\sqrt{2}\Lambda z\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)}{8\,r} \\
& - \frac{\sqrt{2}z e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2}z e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8\,r\sqrt{B(y)}}
\end{aligned}
\tag{80}
$$

Weyl components:

$$
\Psi_0' = \Psi_0 \tag{81}
$$

$$
\begin{aligned}
\Psi_1' = & \frac{1}{4}\Lambda^3 r^2 z A(x)\,B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial x}F(r,x) + \frac{1}{4}i\,\Lambda^3 r^2 z A(x)^{\frac{3}{2}}\sqrt{B(y)}e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial x}F(r,x) \\
& - \frac{1}{4}i\,\Lambda^3 r^2 z\sqrt{A(x)}B(y)^{\frac{3}{2}}\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial y}G(r,y) + \frac{1}{4}\,\Lambda^3 r^2 z A(x)\,B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial y}G(r,y) \\
& - \frac{1}{8}\,\Lambda^2 r^2 z B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)^2 - \frac{1}{8}\,\Lambda^2 r^2 z A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)^2 \\
& - \frac{\Lambda^2 z A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)^2}{4\,r^2} - \frac{\Lambda^2 z B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)^2}{4\,r^2} \\
& + \frac{3\,\Lambda z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial x}F(r,x)}{8\,r^2} + \frac{\Lambda z A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial x)^2}F(r,x)}{4\,r^2} \\
& + \frac{3\,\Lambda z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial y}G(r,y)}{8\,r^2} + \frac{\Lambda z B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial y)^2}G(r,y)}{4\,r^2} \\
& - \frac{z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial x)^2}A(x)}{8\,r^2} - \frac{z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial y)^2}B(y)}{8\,r^2}
\end{aligned}
\tag{82}
$$

$$
\begin{aligned}
\Psi_2' = {} & \frac{1}{4}\,\Lambda^3 r^2 z^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right) + \frac{1}{4} i\,\Lambda^3 r^2 z^2 A\left(x\right)^{\frac{3}{2}} \sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial x} F\left(r,x\right) \\
& - \frac{1}{4} i\,\Lambda^3 r^2 z^2 \sqrt{A\left(x\right)} B\left(y\right)^{\frac{3}{2}} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} G\left(r,y\right) + \frac{1}{4}\,\Lambda^3 r^2 z^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right) \\
& - \frac{1}{8}\,\Lambda^2 r^2 z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 - \frac{1}{8}\,\Lambda^2 r^2 z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 \\
& - \frac{1}{24}\,\Lambda^2 r^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 - \frac{1}{12}\,\Lambda^2 r^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right) \\
& - \frac{1}{24}\,\Lambda^2 r^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 - \frac{1}{12}\,\Lambda^2 r^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right) \\
& - \frac{\Lambda^2 z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{4\,r^2} - \frac{\Lambda^2 z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{4\,r^2} \\
& + \frac{3\,\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{8\,r^2} - \frac{\Lambda^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{12\,r^2} \\
& + \frac{\Lambda z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} + \frac{3\,\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{8\,r^2} \\
& - \frac{\Lambda^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{12\,r^2} + \frac{\Lambda z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} \\
& - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2} + \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{8\,r^2} \\
& + \frac{\Lambda A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{12\,r^2} + \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{8\,r^2} \\
& + \frac{\Lambda B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{12\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{24\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{24\,r^2}
\end{aligned}
\tag{83}
$$

$$
\begin{aligned}
\Psi_3' = {} & \frac{1}{4}\,\Lambda^3 r^2 z^3 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right) + \frac{1}{4} i\,\Lambda^3 r^2 z^3 A\left(x\right)^{\frac{3}{2}} \sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial x} F\left(r,x\right) \\
& - \frac{1}{4} i\,\Lambda^3 r^2 z^3 \sqrt{A\left(x\right)} B\left(y\right)^{\frac{3}{2}} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} G\left(r,y\right) + \frac{1}{4}\,\Lambda^3 r^2 z^3 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right) \\
& - \frac{1}{8}\,\Lambda^2 r^2 z^3 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 - \frac{1}{8}\,\Lambda^2 r^2 z^3 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 \\
& - \frac{1}{8}\,\Lambda^2 r^2 z B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 - \frac{1}{4}\,\Lambda^2 r^2 z A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right) \\
& - \frac{1}{8}\,\Lambda^2 r^2 z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 - \frac{1}{4}\,\Lambda^2 r^2 z A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right) \\
& - \frac{\Lambda^2 z^3 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{4\,r^2} - \frac{\Lambda^2 z^3 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{4\,r^2} \\
& + \frac{3\,\Lambda z^3 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{8\,r^2} - \frac{\Lambda^2 z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{4\,r^2} \\
& + \frac{\Lambda z^3 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} + \frac{3\,\Lambda z^3 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{8\,r^2} \\
& - \frac{\Lambda^2 z B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{4\,r^2} + \frac{\Lambda z^3 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} \\
& - \frac{z^3 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} - \frac{z^3 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2} + \frac{3\,\Lambda z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{8\,r^2} \\
& + \frac{\Lambda z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} + \frac{3\,\Lambda z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{8\,r^2} \\
& + \frac{\Lambda z B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} - \frac{z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} - \frac{z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2}
\end{aligned}
\tag{84}
$$

$$
\begin{aligned}
\Psi_4' = {} & \frac{1}{4}\,\Lambda^3 r^2 z^4 A(x)\,B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial x}F(r,x) + \frac{1}{4}i\,\Lambda^3 r^2 z^4 A(x)^{\frac{3}{2}}\,\sqrt{B(y)}e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial x}F(r,x) \\
& - \frac{1}{4}i\,\Lambda^3 r^2 z^4\sqrt{A(x)}B(y)^{\frac{3}{2}}\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial y}G(r,y) + \frac{1}{4}\,\Lambda^3 r^2 z^4 A(x)\,B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial y}G(r,y) \\
& - \frac{1}{8}\,\Lambda^2 r^2 z^4 B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)^2 - \frac{1}{8}\,\Lambda^2 r^2 z^4 A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)^2 \\
& - \frac{1}{4}\,\Lambda^2 r^2 z^2 B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)^2 - \frac{1}{2}\,\Lambda^2 r^2 z^2 A(x)\,B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial x)^2}A(x) \\
& - \frac{1}{4}\,\Lambda^2 r^2 z^2 A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)^2 - \frac{1}{2}\,\Lambda^2 r^2 z^2 A(x)\,B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial y)^2}B(y) \\
& + \frac{1}{4}\,\Lambda^3 r^2 A(x)\,B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial x}F(r,x) - \frac{1}{4}i\,\Lambda^3 r^2 A(x)^{\frac{3}{2}}\,\sqrt{B(y)}e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial x}F(r,x) \\
& + \frac{1}{4}i\,\Lambda^3 r^2\sqrt{A(x)}B(y)^{\frac{3}{2}}\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial y}G(r,y) + \frac{1}{4}\,\Lambda^3 r^2 A(x)\,B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial y}G(r,y) \\
& - \frac{1}{8}\,\Lambda^2 r^2 B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)^2 - \frac{1}{8}\,\Lambda^2 r^2 A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)^2 \\
& - \frac{\Lambda^2 z^4 A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)^2}{4\,r^2} - \frac{\Lambda^2 z^4 B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)^2}{4\,r^2} \\
& + \frac{3\,\Lambda z^4 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial x}F(r,x)}{8\,r^2} - \frac{\Lambda^2 z^2 A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)^2}{2\,r^2} \\
& + \frac{\Lambda z^4 A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial x)^2}F(r,x)}{4\,r^2} + \frac{3\,\Lambda z^4 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial y}G(r,y)}{8\,r^2} \\
& - \frac{\Lambda^2 z^2 B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)^2}{2\,r^2} + \frac{\Lambda z^4 B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial y)^2}G(r,y)}{4\,r^2} \\
& - \frac{z^4 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial x)^2}A(x)}{8\,r^2} - \frac{z^4 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial y)^2}B(y)}{8\,r^2} + \frac{3\,\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial x}F(r,x)}{4\,r^2} \\
& - \frac{\Lambda^2 A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)^2}{4\,r^2} + \frac{\Lambda z^2 A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial x)^2}F(r,x)}{2\,r^2} \\
& + \frac{3\,\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial y}G(r,y)}{4\,r^2} - \frac{\Lambda^2 B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)^2}{4\,r^2} \\
& + \frac{\Lambda z^2 B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial y)^2}G(r,y)}{2\,r^2} - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial x)^2}A(x)}{4\,r^2} \\
& - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial y)^2}B(y)}{4\,r^2} + \frac{3\,\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial x}F(r,x)}{8\,r^2} \\
& + \frac{\Lambda A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial x)^2}F(r,x)}{4\,r^2} + \frac{3\,\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial y}G(r,y)}{24\qquad 8\,r^2} \\
& + \frac{\Lambda B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial y)^2}G(r,y)}{4\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial x)^2}A(x)}{8\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial y)^2}B(y)}{8\,r^2}
\end{aligned}
$$

Ricci components:

$$\Phi'_{00} = \Phi_{00} \tag{86}$$

$$\begin{aligned}\Phi'_{01} = {} & \frac{\Lambda z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) \frac{\partial}{\partial x} F(r,x)}{4\,r^2} + \frac{\Lambda z A(x)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F(r,x)}{4\,r^2} \\ & - \frac{\Lambda z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \frac{\partial}{\partial y} G(r,y)}{4\,r^2} - \frac{\Lambda z B(y)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G(r,y)}{4\,r^2} \\ & - \frac{z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A(x)}{8\,r^2} + \frac{z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B(y)}{8\,r^2}\end{aligned} \tag{87}$$

$$\begin{aligned}\Phi'_{02} = {} & -\frac{1}{8}\,\Lambda^2 r^2 B(y)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)^2 - \frac{1}{4} i\,\Lambda^2 r^2 \sqrt{A(x)}\sqrt{B(y)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) \frac{\partial}{\partial y} B(y) \\ & + \frac{1}{8}\,\Lambda^2 r^2 A(x)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)^2 + \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) \frac{\partial}{\partial x} F(r,x)}{4\,r^2} \\ & - \frac{\Lambda^2 A(x)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)^2}{8\,r^2} + \frac{\Lambda z^2 A(x)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F(r,x)}{4\,r^2} \\ & - \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \frac{\partial}{\partial y} G(r,y)}{4\,r^2} + \frac{i\,\Lambda^2 \sqrt{A(x)}\sqrt{B(y)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x) \frac{\partial}{\partial y} G(r,y)}{4\,r^2} \\ & + \frac{\Lambda^2 B(y)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)^2}{8\,r^2} - \frac{\Lambda z^2 B(y)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G(r,y)}{4\,r^2} \\ & - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A(x)}{8\,r^2} + \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B(y)}{8\,r^2} + \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) \frac{\partial}{\partial x} F(r,x)}{4\,r^2} \\ & - \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \frac{\partial}{\partial y} G(r,y)}{4\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A(x)}{8\,r^2} + \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B(y)}{8\,r^2}\end{aligned} \tag{88}$$

$$
\begin{aligned}
\Phi'_{10} = & \frac{\Lambda z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) \frac{\partial}{\partial x} F(r,x)}{4\, r^2} + \frac{\Lambda z A(x)\, e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F(r,x)}{4\, r^2} \\
& - \frac{\Lambda z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \frac{\partial}{\partial y} G(r,y)}{4\, r^2} - \frac{\Lambda z B(y)\, e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G(r,y)}{4\, r^2} \\
& - \frac{z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A(x)}{8\, r^2} + \frac{z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B(y)}{8\, r^2}
\end{aligned} \tag{89}
$$

$$
\begin{aligned}
\Phi'_{11} = & \frac{1}{16} \Lambda^2 r^2 B(y)\, e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)^2 + \frac{1}{8} \Lambda^2 r^2 A(x)\, B(y)\, e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A(x) \\
& + \frac{1}{16} \Lambda^2 r^2 A(x)\, e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)^2 + \frac{1}{8} \Lambda^2 r^2 A(x)\, B(y)\, e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B(y) \\
& + \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) \frac{\partial}{\partial x} F(r,x)}{4\, r^2} - \frac{\Lambda^2 A(x)\, e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)^2}{16\, r^2} \\
& + \frac{\Lambda z^2 A(x)\, e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F(r,x)}{4\, r^2} - \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \frac{\partial}{\partial y} G(r,y)}{4\, r^2} \\
& - \frac{\Lambda^2 B(y)\, e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)^2}{16\, r^2} - \frac{\Lambda z^2 B(y)\, e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G(r,y)}{4\, r^2} \\
& - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A(x)}{8\, r^2} + \frac{z^2 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B(y)}{8\, r^2} \\
& - \frac{\Lambda A(x)\, e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F(r,x)}{8\, r^2} - \frac{\Lambda B(y)\, e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G(r,y)}{8\, r^2}
\end{aligned} \tag{90}
$$

$$
\begin{aligned}
\Phi'_{12} = {} & \frac{1}{4}\,\Lambda^2 r^2 z A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial x\right)^2} A\left(x\right) - \frac{1}{4} i\,\Lambda^2 r^2 z \sqrt{A\left(x\right)} \sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} B\left(y\right) \\
& + \frac{1}{4}\,\Lambda^2 r^2 z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 + \frac{1}{4}\,\Lambda^2 r^2 z A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial y\right)^2} B\left(y\right) \\
& + \frac{\Lambda z^3 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} - \frac{\Lambda^2 z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{4\,r^2} \\
& + \frac{\Lambda z^3 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial x\right)^2} F\left(r,x\right)}{4\,r^2} - \frac{\Lambda z^3 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} \\
& + \frac{i\,\Lambda^2 z \sqrt{A\left(x\right)} \sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} - \frac{\Lambda z^3 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial y\right)^2} G\left(r,y\right)}{4\,r^2} \\
& - \frac{z^3 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial x\right)^2} A\left(x\right)}{8\,r^2} + \frac{z^3 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial y\right)^2} B\left(y\right)}{8\,r^2} + \frac{\Lambda z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} \\
& - \frac{\Lambda z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial x\right)^2} F\left(r,x\right)}{4\,r^2} - \frac{\Lambda z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} \\
& - \frac{\Lambda z B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial y\right)^2} G\left(r,y\right)}{4\,r^2} - \frac{z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial x\right)^2} A\left(x\right)}{8\,r^2} + \frac{z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial y\right)^2} B\left(y\right)}{8\,r^2}
\end{aligned}
\tag{91}
$$

$$
\begin{aligned}
\Phi'_{20} = & -\frac{1}{8}\,\Lambda^2 r^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 + \frac{1}{4} i\,\Lambda^2 r^2 \sqrt{A\left(x\right)}\sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} B\left(y\right) \\
& + \frac{1}{8}\,\Lambda^2 r^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 + \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} \\
& - \frac{\Lambda^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{8\,r^2} + \frac{\Lambda z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} \\
& - \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} - \frac{i\,\Lambda^2 \sqrt{A\left(x\right)}\sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} \\
& + \frac{\Lambda^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{8\,r^2} - \frac{\Lambda z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} \\
& - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} + \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2} + \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} \\
& - \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} + \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2}
\end{aligned}
\tag{92}
$$

$$
\begin{aligned}
\Phi'_{21} = {} & \frac{1}{4}\,\Lambda^2 r^2 z A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial x\right)^2} A\left(x\right) + \frac{1}{4} i\,\Lambda^2 r^2 z \sqrt{A\left(x\right)} \sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} B\left(y\right) \\
& + \frac{1}{4}\,\Lambda^2 r^2 z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 + \frac{1}{4}\,\Lambda^2 r^2 z A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial y\right)^2} B\left(y\right) \\
& + \frac{\Lambda z^3 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} - \frac{\Lambda^2 z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{4\,r^2} \\
& + \frac{\Lambda z^3 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial x\right)^2} F\left(r,x\right)}{4\,r^2} - \frac{\Lambda z^3 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} \\
& - \frac{i\,\Lambda^2 z \sqrt{A\left(x\right)} \sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} - \frac{\Lambda z^3 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial y\right)^2} G\left(r,y\right)}{4\,r^2} \\
& - \frac{z^3 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial x\right)^2} A\left(x\right)}{8\,r^2} + \frac{z^3 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial y\right)^2} B\left(y\right)}{8\,r^2} + \frac{\Lambda z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} \\
& - \frac{\Lambda z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial x\right)^2} F\left(r,x\right)}{4\,r^2} - \frac{\Lambda z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} \\
& - \frac{\Lambda z B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial y\right)^2} G\left(r,y\right)}{4\,r^2} - \frac{z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial x\right)^2} A\left(x\right)}{8\,r^2} + \frac{z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial y\right)^2} B\left(y\right)}{8\,r^2}
\end{aligned}
\tag{93}
$$

$$
\begin{aligned}
\Phi'_{22} = {} & \frac{1}{2}\,\Lambda^2 r^2 z^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right) + \frac{1}{2}\,\Lambda^2 r^2 z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 \\
& + \frac{1}{2}\,\Lambda^2 r^2 z^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right) + \frac{\Lambda z^4 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} \\
& - \frac{\Lambda^2 z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{2\,r^2} + \frac{\Lambda z^4 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} \\
& - \frac{\Lambda z^4 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} - \frac{\Lambda z^4 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} \\
& - \frac{z^4 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} + \frac{z^4 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2} + \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{2\,r^2} \\
& - \frac{\Lambda z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{2\,r^2} - \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{2\,r^2} \\
& - \frac{\Lambda z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{2\,r^2} - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{4\,r^2} \\
& + \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{4\,r^2} + \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} \\
& + \frac{\Lambda A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} - \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} \\
& - \frac{\Lambda B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} + \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2}
\end{aligned}
\tag{94}
$$

B Type transformations:

Spin coefficients:

$$\rho' = 0 \tag{95}$$

$$\sigma' = 0 \tag{96}$$

$$\kappa' = -\frac{\sqrt{2}\Lambda z^3\sqrt{A(x)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)}{4r} + \frac{i\sqrt{2}\Lambda z^3\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)}{4r} + \frac{\sqrt{2}z^3 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8r\sqrt{A(x)}} - \frac{i\sqrt{2}z^3 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8r\sqrt{B(y)}} \tag{97}$$

$$\mu' = 0 \tag{98}$$

$$\lambda' = 0 \tag{99}$$

$$\nu' = \frac{\sqrt{2}\Lambda\sqrt{A(x)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)}{4rz^3} + \frac{i\sqrt{2}\Lambda\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)}{4rz^3} - \frac{\sqrt{2}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8rz^3\sqrt{A(x)}} - \frac{i\sqrt{2}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8rz^3\sqrt{B(y)}} \tag{100}$$

$$\alpha' = \alpha \tag{101}$$

$$\beta' = \beta \tag{102}$$

$$\tau' = -\frac{\sqrt{2}z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8r\sqrt{A(x)}} - \frac{i\sqrt{2}z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8r\sqrt{B(y)}} \tag{103}$$

$$\pi' = \pi \tag{104}$$

$$\epsilon' = 0 \tag{105}$$

$$\gamma' = 0 \tag{106}$$

Weyl components:

$$\begin{aligned}
\Psi_0' = {} & \frac{1}{4}\,\Lambda^3 r^2 z^4 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right) + \frac{1}{4} i\,\Lambda^3 r^2 z^4 A\left(x\right)^{\frac{3}{2}} \sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial x} F\left(r,x\right) \\
& - \frac{1}{4} i\,\Lambda^3 r^2 z^4 \sqrt{A\left(x\right)} B\left(y\right)^{\frac{3}{2}} e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} G\left(r,y\right) + \frac{1}{4}\,\Lambda^3 r^2 z^4 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right) \\
& - \frac{1}{8}\,\Lambda^2 r^2 z^4 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 - \frac{1}{8}\,\Lambda^2 r^2 z^4 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 \\
& - \frac{\Lambda^2 z^4 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{4\,r^2} - \frac{\Lambda^2 z^4 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{4\,r^2} \\
& + \frac{3\,\Lambda z^4 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{8\,r^2} + \frac{\Lambda z^4 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} \\
& + \frac{3\,\Lambda z^4 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{8\,r^2} + \frac{\Lambda z^4 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} \\
& - \frac{z^4 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} - \frac{z^4 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2}
\end{aligned} \tag{107}$$

$$\Psi_1' = 0 \tag{108}$$

$$
\begin{aligned}
\Psi_2' = & -\frac{1}{24}\,\Lambda^2 r^2 z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 - \frac{1}{12}\,\Lambda^2 r^2 z^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right) \\
& - \frac{1}{24}\,\Lambda^2 r^2 z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 - \frac{1}{12}\,\Lambda^2 r^2 z^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right) \\
& - \frac{\Lambda^2 z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{12\,r^2} - \frac{\Lambda^2 z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{12\,r^2} \\
& + \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{8\,r^2} + \frac{\Lambda z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{12\,r^2} \\
& + \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{8\,r^2} + \frac{\Lambda z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{12\,r^2} \\
& - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{24\,r^2} - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{24\,r^2}
\end{aligned}
\tag{109}
$$

$$
\Psi_3' = 0 \tag{110}
$$

$$
\Psi_4' = \Psi_4 \tag{111}
$$

Ricci components:

$$
\begin{aligned}
\Phi_{00}' = & \frac{\Lambda z^4 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} + \frac{\Lambda z^4 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} \\
& - \frac{\Lambda z^4 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} - \frac{\Lambda z^4 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} \\
& - \frac{z^4 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} + \frac{z^4 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2}
\end{aligned}
\tag{112}
$$

$$
\Phi_{01}' = 0 \tag{113}
$$

$$
\begin{aligned}
\Phi'_{02} = & -\frac{1}{8}\,\Lambda^2 r^2 z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 - \frac{1}{4} i\,\Lambda^2 r^2 z^2 \sqrt{A\left(x\right)}\sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} B\left(y\right) \\
& + \frac{1}{8}\,\Lambda^2 r^2 z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 - \frac{\Lambda^2 z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{8\,r^2} \\
& + \frac{i\,\Lambda^2 z^2 \sqrt{A\left(x\right)}\sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} + \frac{\Lambda^2 z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{8\,r^2} \\
& + \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} - \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} \\
& - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} + \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2}
\end{aligned}
\tag{114}
$$

$$
\Phi'_{10} = 0 \tag{115}
$$

$$
\begin{aligned}
\Phi'_{11} = & \frac{1}{16}\,\Lambda^2 r^2 z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 + \frac{1}{8}\,\Lambda^2 r^2 z^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right) \\
& + \frac{1}{16}\,\Lambda^2 r^2 z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 + \frac{1}{8}\,\Lambda^2 r^2 z^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right) \\
& - \frac{\Lambda^2 z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{16\,r^2} - \frac{\Lambda^2 z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{16\,r^2} \\
& - \frac{\Lambda z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{8\,r^2} - \frac{\Lambda z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{8\,r^2}
\end{aligned}
\tag{116}
$$

$$
\Phi'_{12} = 0 \tag{117}
$$

$$
\begin{aligned}
\Phi'_{20} = & -\frac{1}{8}\,\Lambda^2 r^2 z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 + \frac{1}{4} i\,\Lambda^2 r^2 z^2 \sqrt{A\left(x\right)}\sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} B\left(y\right) \\
& + \frac{1}{8}\,\Lambda^2 r^2 z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 - \frac{\Lambda^2 z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{8\,r^2} \\
& - \frac{i\,\Lambda^2 z^2 \sqrt{A\left(x\right)}\sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} + \frac{\Lambda^2 z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{8\,r^2} \\
& + \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} - \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} \\
& - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} + \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2}
\end{aligned}
\tag{118}
$$

$$
\Phi'_{21} = 0 \tag{119}
$$

$$
\Phi'_{22} = \Phi_{22} \tag{120}
$$

C Type transformations:

Spin coefficients:

$$\rho' = \frac{\sqrt{2}\Lambda z^2 \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{4\,r} + \frac{i\sqrt{2}\Lambda z^2 \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{4\,r}$$
$$- \frac{\sqrt{2} z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2} z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}} \tag{121}$$

$$\sigma' = -\frac{1}{4}\sqrt{2}\Lambda^2 r^3 z \sqrt{A(x)} B(y)\, e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) - \frac{1}{4} i\sqrt{2}\Lambda^2 r^3 z A(x) \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)$$
$$+ \frac{\sqrt{2}\Lambda z^3 \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{4\,r} + \frac{i\sqrt{2}\Lambda z^3 \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{4\,r}$$
$$- \frac{\sqrt{2} z^3 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2} z^3 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}} \tag{122}$$
$$+ \frac{\sqrt{2}\Lambda z \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{4\,r} + \frac{i\sqrt{2}\Lambda z \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{4\,r}$$
$$- \frac{\sqrt{2} z e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2} z e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}}$$

$$
\begin{aligned}
\kappa' = {} & \frac{1}{8}\sqrt{2}\Lambda^2 r^3 z^2 \sqrt{A(x)} B(y)\, e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) - \frac{3}{8} i \sqrt{2}\Lambda^2 r^3 z^2 A(x) \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \\
& + \frac{\sqrt{2}\Lambda z^4 \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{4r} + \frac{i\sqrt{2}\Lambda z^4 \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{4r} \\
& - \frac{\sqrt{2} z^4 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8r\sqrt{A(x)}} - \frac{i\sqrt{2} z^4 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8r\sqrt{B(y)}} \\
& - \frac{\sqrt{2}\Lambda z^2 \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{8r} + \frac{3i\sqrt{2}\Lambda z^2 \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{8r} \\
& + \frac{\sqrt{2} z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8r\sqrt{A(x)}} - \frac{i\sqrt{2} z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8r\sqrt{B(y)}} \\
& - \frac{\sqrt{2}\Lambda \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{4r} + \frac{i\sqrt{2}\Lambda \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{4r} \\
& + \frac{\sqrt{2} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8r\sqrt{A(x)}} - \frac{i\sqrt{2} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8r\sqrt{B(y)}}
\end{aligned}
\tag{123}
$$

$$
\begin{aligned}
\mu' = {} & \frac{1}{4}\sqrt{2}\Lambda^2 r^3 z \sqrt{A(x)} B(y)\, e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) - \frac{1}{4} i \sqrt{2}\Lambda^2 r^3 z A(x) \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \\
& + \frac{\sqrt{2}\Lambda z^3 \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{4r} + \frac{i\sqrt{2}\Lambda z^3 \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{4r} \\
& - \frac{\sqrt{2} z^3 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8r\sqrt{A(x)}} - \frac{i\sqrt{2} z^3 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8r\sqrt{B(y)}} \\
& - \frac{\sqrt{2}\Lambda z \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{4r} + \frac{i\sqrt{2}\Lambda z \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{4r} \\
& - \frac{\sqrt{2} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{4r\sqrt{A(x)}} - \frac{i\sqrt{2} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{4r\sqrt{B(y)}}
\end{aligned}
\tag{124}
$$

$$\lambda' = \frac{\sqrt{2}\Lambda z\sqrt{A(x)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)}{4r} + \frac{i\sqrt{2}\Lambda z\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)}{4r}$$
$$- \frac{\sqrt{2}ze^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8r\sqrt{A(x)}} - \frac{i\sqrt{2}ze^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8r\sqrt{B(y)}} \tag{125}$$

$$\nu' = \nu \tag{126}$$

$$\alpha' = \frac{1}{8}\sqrt{2}\Lambda^2 r^3\sqrt{A(x)}B(y)\,e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x) - \frac{1}{8}i\sqrt{2}\Lambda^2 r^3 A(x)\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)$$
$$+ \frac{\sqrt{2}\Lambda z^2\sqrt{A(x)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)}{4r} + \frac{i\sqrt{2}\Lambda z^2\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)}{4r}$$
$$- \frac{\sqrt{2}z^2e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8r\sqrt{A(x)}} - \frac{i\sqrt{2}z^2e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8r\sqrt{B(y)}} \tag{127}$$
$$- \frac{\sqrt{2}\Lambda\sqrt{A(x)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)}{8r} + \frac{i\sqrt{2}\Lambda\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)}{8r}$$
$$+ \frac{\sqrt{2}ze^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8r\sqrt{A(x)}} - \frac{i\sqrt{2}ze^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8r\sqrt{B(y)}}$$

$$\beta' = -\frac{1}{8}\sqrt{2}\Lambda^2 r^3\sqrt{A(x)}B(y)\,e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x) - \frac{1}{8}i\sqrt{2}\Lambda^2 r^3 A(x)\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)$$
$$+ \frac{\sqrt{2}\Lambda z^2\sqrt{A(x)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)}{4r} + \frac{i\sqrt{2}\Lambda z^2\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)}{4r}$$
$$- \frac{\sqrt{2}z^2e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)}{8r\sqrt{A(x)}} - \frac{i\sqrt{2}z^2e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)}{8r\sqrt{B(y)}} \tag{128}$$
$$+ \frac{\sqrt{2}\Lambda\sqrt{A(x)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)}{8r} + \frac{i\sqrt{2}\Lambda\sqrt{B(y)}e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y)-\frac{1}{2}\Lambda F(r,x)-\frac{1}{2}\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)}{8r}$$

$$\tau' = \frac{\sqrt{2}\Lambda z^2 \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{4\,r} + \frac{i\sqrt{2}\Lambda z^2 \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{4\,r}$$
$$- \frac{\sqrt{2} z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2} z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}}$$
$$- \frac{\sqrt{2} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}} \tag{129}$$

$$\pi' = \frac{\sqrt{2}\Lambda z^2 \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{4\,r} + \frac{i\sqrt{2}\Lambda z^2 \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{4\,r}$$
$$- \frac{\sqrt{2} z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2} z^2 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}}$$
$$+ \frac{\sqrt{2} z e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{4\,r\sqrt{A(x)}} - \frac{i\sqrt{2} z e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{4\,r\sqrt{B(y)}}$$
$$+ \frac{\sqrt{2} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}} \tag{130}$$

$$
\begin{aligned}
\epsilon' = & -\frac{1}{8}\sqrt{2}\Lambda^2 r^3 z^2 \sqrt{A(x)} B(y)\, e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) - \frac{1}{8} i \sqrt{2}\Lambda^2 r^3 z^2 A(x) \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \\
& + \frac{1}{8}\sqrt{2}\Lambda^2 r^3 z \sqrt{A(x)} B(y)\, e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) - \frac{1}{8} i \sqrt{2}\Lambda^2 r^3 z A(x) \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \\
& + \frac{\sqrt{2}\Lambda z^3 \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{4\,r} + \frac{i\sqrt{2}\Lambda z^3 \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{4\,r} \\
& + \frac{\sqrt{2}\Lambda z^2 \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{8\,r} + \frac{i\sqrt{2}\Lambda z^2 \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{8\,r} \\
& - \frac{\sqrt{2} z^3 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2} z^3 e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}} \\
& - \frac{\sqrt{2}\Lambda z \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{8\,r} + \frac{i\sqrt{2}\Lambda z \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{8\,r} \\
& + \frac{\sqrt{2} z e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2} z e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}}
\end{aligned}
\tag{131}
$$

$$
\begin{aligned}
\gamma' = & \frac{\sqrt{2}\Lambda z \sqrt{A(x)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)}{4\,r} + \frac{i\sqrt{2}\Lambda z \sqrt{B(y)} e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)}{4\,r} \\
& - \frac{\sqrt{2} z e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)}{8\,r\sqrt{A(x)}} - \frac{i\sqrt{2} z e^{\left(\frac{1}{2}\Lambda^2 r^4 A(x)B(y) - \frac{1}{2}\Lambda F(r,x) - \frac{1}{2}\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)}{8\,r\sqrt{B(y)}}
\end{aligned}
\tag{132}
$$

Weyl components:

$$
\begin{aligned}
\Psi_0' = {} & \frac{1}{4}\,\Lambda^3 r^2 z^4 A(x)\,B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial x}F(r,x) - \frac{1}{4}i\,\Lambda^3 r^2 z^4 A(x)^{\frac{3}{2}}\,\sqrt{B(y)}e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial x}F(r,x) \\
& + \frac{1}{4}i\,\Lambda^3 r^2 z^4\sqrt{A(x)}B(y)^{\frac{3}{2}}\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial y}G(r,y) + \frac{1}{4}\,\Lambda^3 r^2 z^4 A(x)\,B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial y}G(r,y) \\
& - \frac{1}{8}\,\Lambda^2 r^2 z^4 B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)^2 - \frac{1}{8}\,\Lambda^2 r^2 z^4 A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)^2 \\
& - \frac{1}{4}\,\Lambda^2 r^2 z^2 B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)^2 - \frac{1}{2}\,\Lambda^2 r^2 z^2 A(x)\,B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial x)^2}A(x) \\
& - \frac{1}{4}\,\Lambda^2 r^2 z^2 A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)^2 - \frac{1}{2}\,\Lambda^2 r^2 z^2 A(x)\,B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial y)^2}B(y) \\
& + \frac{1}{4}\,\Lambda^3 r^2 A(x)\,B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial x}F(r,x) + \frac{1}{4}i\,\Lambda^3 r^2 A(x)^{\frac{3}{2}}\,\sqrt{B(y)}e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial x}F(r,x) \\
& - \frac{1}{4}i\,\Lambda^3 r^2\sqrt{A(x)}B(y)^{\frac{3}{2}}\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial y}G(r,y) + \frac{1}{4}\,\Lambda^3 r^2 A(x)\,B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial y}G(r,y) \\
& - \frac{1}{8}\,\Lambda^2 r^2 B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)^2 - \frac{1}{8}\,\Lambda^2 r^2 A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)^2 \\
& - \frac{\Lambda^2 z^4 A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)^2}{4\,r^2} - \frac{\Lambda^2 z^4 B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)^2}{4\,r^2} \\
& + \frac{3\,\Lambda z^4 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial x}F(r,x)}{8\,r^2} - \frac{\Lambda^2 z^2 A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)^2}{2\,r^2} \\
& + \frac{\Lambda z^4 A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial x)^2}F(r,x)}{4\,r^2} + \frac{3\,\Lambda z^4 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial y}G(r,y)}{8\,r^2} \\
& - \frac{\Lambda^2 z^2 B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)^2}{2\,r^2} + \frac{\Lambda z^4 B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial y)^2}G(r,y)}{4\,r^2} \\
& - \frac{z^4 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial x)^2}A(x)}{8\,r^2} - \frac{z^4 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial y)^2}B(y)}{8\,r^2} + \frac{3\,\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial x}F(r,x)}{4\,r^2} \\
& - \frac{\Lambda^2 A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}F(r,x)^2}{4\,r^2} + \frac{\Lambda z^2 A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial x)^2}F(r,x)}{2\,r^2} \\
& + \frac{3\,\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial y}G(r,y)}{4\,r^2} - \frac{\Lambda^2 B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}G(r,y)^2}{4\,r^2} \\
& + \frac{\Lambda z^2 B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial y)^2}G(r,y)}{2\,r^2} - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial x)^2}A(x)}{4\,r^2} \\
& - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial y)^2}B(y)}{4\,r^2} + \frac{3\,\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial x}A(x)\,\frac{\partial}{\partial x}F(r,x)}{8\,r^2} \\
& + \frac{\Lambda A(x)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial x)^2}F(r,x)}{4\,r^2} + \frac{3\,\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial}{\partial y}B(y)\,\frac{\partial}{\partial y}G(r,y)}{41\qquad 8\,r^2} \\
& + \frac{\Lambda B(y)\,e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial y)^2}G(r,y)}{4\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial x)^2}A(x)}{8\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)}\frac{\partial^2}{(\partial y)^2}B(y)}{8\,r^2}
\end{aligned}
$$

$$
\begin{aligned}
\Psi_1' = {} & \frac{1}{4}\,\Lambda^3 r^2 z^3 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right) - \frac{1}{4} i\,\Lambda^3 r^2 z^3 A\left(x\right)^{\frac{3}{2}} \sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial x} F\left(r,x\right) \\
& + \frac{1}{4} i\,\Lambda^3 r^2 z^3 \sqrt{A\left(x\right)} B\left(y\right)^{\frac{3}{2}} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} G\left(r,y\right) + \frac{1}{4}\,\Lambda^3 r^2 z^3 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right) \\
& - \frac{1}{8}\,\Lambda^2 r^2 z^3 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 - \frac{1}{8}\,\Lambda^2 r^2 z^3 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 \\
& - \frac{1}{8}\,\Lambda^2 r^2 z B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 - \frac{1}{4}\,\Lambda^2 r^2 z A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right) \\
& - \frac{1}{8}\,\Lambda^2 r^2 z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 - \frac{1}{4}\,\Lambda^2 r^2 z A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right) \\
& - \frac{\Lambda^2 z^3 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{4\,r^2} - \frac{\Lambda^2 z^3 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{4\,r^2} \\
& + \frac{3\,\Lambda z^3 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{8\,r^2} - \frac{\Lambda^2 z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{4\,r^2} \\
& + \frac{\Lambda z^3 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} + \frac{3\,\Lambda z^3 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{8\,r^2} \\
& - \frac{\Lambda^2 z B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{4\,r^2} + \frac{\Lambda z^3 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} \\
& - \frac{z^3 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} - \frac{z^3 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2} + \frac{3\,\Lambda z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{8\,r^2} \\
& + \frac{\Lambda z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} + \frac{3\,\Lambda z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{8\,r^2} \\
& + \frac{\Lambda z B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} - \frac{z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} - \frac{z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2}
\end{aligned}
\tag{134}
$$

$$
\begin{aligned}
\Psi_2' = {} & \frac{1}{4}\,\Lambda^3 r^2 z^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right) - \frac{1}{4} i\,\Lambda^3 r^2 z^2 A\left(x\right)^{\frac{3}{2}} \sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial x} F\left(r,x\right) \\
& + \frac{1}{4} i\,\Lambda^3 r^2 z^2 \sqrt{A\left(x\right)} B\left(y\right)^{\frac{3}{2}} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} G\left(r,y\right) + \frac{1}{4}\,\Lambda^3 r^2 z^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right) \\
& - \frac{1}{8}\,\Lambda^2 r^2 z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 - \frac{1}{8}\,\Lambda^2 r^2 z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 \\
& - \frac{1}{24}\,\Lambda^2 r^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 - \frac{1}{12}\,\Lambda^2 r^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right) \\
& - \frac{1}{24}\,\Lambda^2 r^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 - \frac{1}{12}\,\Lambda^2 r^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right) \\
& - \frac{\Lambda^2 z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{4\,r^2} - \frac{\Lambda^2 z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{4\,r^2} \\
& + \frac{3\,\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{8\,r^2} - \frac{\Lambda^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{12\,r^2} \\
& + \frac{\Lambda z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} + \frac{3\,\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{8\,r^2} \\
& - \frac{\Lambda^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{12\,r^2} + \frac{\Lambda z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} \\
& - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2} + \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{8\,r^2} \\
& + \frac{\Lambda A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{12\,r^2} + \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{8\,r^2} \\
& + \frac{\Lambda B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{12\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{24\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{24\,r^2}
\end{aligned}
\tag{135}
$$

$$
\begin{aligned}
\Psi_3' = {} & \frac{1}{4}\,\Lambda^3 r^2 z A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right) - \frac{1}{4} i\,\Lambda^3 r^2 z A\left(x\right)^{\frac{3}{2}} \sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial x} F\left(r,x\right) \\
& + \frac{1}{4} i\,\Lambda^3 r^2 z \sqrt{A\left(x\right)} B\left(y\right)^{\frac{3}{2}} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} G\left(r,y\right) + \frac{1}{4}\,\Lambda^3 r^2 z A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right) \\
& - \frac{1}{8}\,\Lambda^2 r^2 z B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 - \frac{1}{8}\,\Lambda^2 r^2 z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 \\
& - \frac{\Lambda^2 z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{4\,r^2} - \frac{\Lambda^2 z B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{4\,r^2} \\
& + \frac{3\,\Lambda z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{8\,r^2} + \frac{\Lambda z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} \\
& + \frac{3\,\Lambda z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{8\,r^2} + \frac{\Lambda z B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} \\
& - \frac{z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} - \frac{z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2}
\end{aligned}
\tag{136}
$$

$$
\Psi_4' = \Psi_4 \tag{137}
$$

Ricci components:

$$
\begin{aligned}
\Phi'_{00} = {} & \frac{1}{2}\,\Lambda^2 r^2 z^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right) + \frac{1}{2}\,\Lambda^2 r^2 z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 \\
& + \frac{1}{2}\,\Lambda^2 r^2 z^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right) + \frac{\Lambda z^4 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} \\
& - \frac{\Lambda^2 z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{2\,r^2} + \frac{\Lambda z^4 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} \\
& - \frac{\Lambda z^4 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} - \frac{\Lambda z^4 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} \\
& - \frac{z^4 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} + \frac{z^4 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2} + \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{2\,r^2} \\
& - \frac{\Lambda z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{2\,r^2} - \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{2\,r^2} \\
& - \frac{\Lambda z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{2\,r^2} - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{4\,r^2} \\
& + \frac{z^2 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{4\,r^2} + \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} \\
& + \frac{\Lambda A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} - \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} \\
& - \frac{\Lambda B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} + \frac{e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2}
\end{aligned}
\tag{138}
$$

$$
\begin{aligned}
\Phi'_{01} = {} & \frac{1}{4}\,\Lambda^2 r^2 z A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial x\right)^2} A\left(x\right) - \frac{1}{4} i\,\Lambda^2 r^2 z \sqrt{A\left(x\right)} \sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} B\left(y\right) \\
& + \frac{1}{4}\,\Lambda^2 r^2 z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 + \frac{1}{4}\,\Lambda^2 r^2 z A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial y\right)^2} B\left(y\right) \\
& + \frac{\Lambda z^3 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} - \frac{\Lambda^2 z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{4\,r^2} \\
& + \frac{\Lambda z^3 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial x\right)^2} F\left(r,x\right)}{4\,r^2} - \frac{\Lambda z^3 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} \\
& + \frac{i\,\Lambda^2 z \sqrt{A\left(x\right)} \sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} - \frac{\Lambda z^3 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial y\right)^2} G\left(r,y\right)}{4\,r^2} \\
& - \frac{z^3 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial x\right)^2} A\left(x\right)}{8\,r^2} + \frac{z^3 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial y\right)^2} B\left(y\right)}{8\,r^2} + \frac{\Lambda z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} \\
& - \frac{\Lambda z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial x\right)^2} F\left(r,x\right)}{4\,r^2} - \frac{\Lambda z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} \\
& - \frac{\Lambda z B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial y\right)^2} G\left(r,y\right)}{4\,r^2} - \frac{z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial x\right)^2} A\left(x\right)}{8\,r^2} + \frac{z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{\left(\partial y\right)^2} B\left(y\right)}{8\,r^2}
\end{aligned}
\tag{139}
$$

$$
\begin{aligned}
\Phi'_{02} = & -\frac{1}{8}\,\Lambda^2 r^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 - \frac{1}{4} i\,\Lambda^2 r^2 \sqrt{A\left(x\right)}\sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} B\left(y\right) \\
& + \frac{1}{8}\,\Lambda^2 r^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 + \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} \\
& - \frac{\Lambda^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{8\,r^2} + \frac{\Lambda z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\,r^2} \\
& - \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} + \frac{i\,\Lambda^2 \sqrt{A\left(x\right)}\sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} \\
& + \frac{\Lambda^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{8\,r^2} - \frac{\Lambda z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\,r^2} \\
& - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} + \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2} + \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\,r^2} \\
& - \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\,r^2} + \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\,r^2}
\end{aligned}
\tag{140}
$$

$$
\begin{aligned}
\Phi'_{10} = {} & \frac{1}{4}\,\Lambda^2 r^2 z A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right) + \frac{1}{4}\, i\, \Lambda^2 r^2 z \sqrt{A\left(x\right)} \sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial y} B\left(y\right) \\
& + \frac{1}{4}\,\Lambda^2 r^2 z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 + \frac{1}{4}\,\Lambda^2 r^2 z A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right) \\
& + \frac{\Lambda z^3 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\, r^2} - \frac{\Lambda^2 z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{4\, r^2} \\
& + \frac{\Lambda z^3 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\, r^2} - \frac{\Lambda z^3 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\, r^2} \\
& - \frac{i\, \Lambda^2 z \sqrt{A\left(x\right)} \sqrt{B\left(y\right)} e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\, r^2} - \frac{\Lambda z^3 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\, r^2} \\
& - \frac{z^3 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\, r^2} + \frac{z^3 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\, r^2} + \frac{\Lambda z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\, r^2} \\
& - \frac{\Lambda z A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\, r^2} - \frac{\Lambda z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\, r^2} \\
& - \frac{\Lambda z B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\, r^2} - \frac{z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\, r^2} + \frac{z e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\, r^2}
\end{aligned}
\tag{141}
$$

$$
\begin{aligned}
\Phi'_{11} = {} & \frac{1}{16}\,\Lambda^2 r^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right)^2 + \frac{1}{8}\,\Lambda^2 r^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right) \\
& + \frac{1}{16}\,\Lambda^2 r^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right)^2 + \frac{1}{8}\,\Lambda^2 r^2 A\left(x\right) B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right) \\
& + \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} A\left(x\right) \frac{\partial}{\partial x} F\left(r,x\right)}{4\, r^2} - \frac{\Lambda^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial x} F\left(r,x\right)^2}{16\, r^2} \\
& + \frac{\Lambda z^2 A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{4\, r^2} - \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} B\left(y\right) \frac{\partial}{\partial y} G\left(r,y\right)}{4\, r^2} \\
& - \frac{\Lambda^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial}{\partial y} G\left(r,y\right)^2}{16\, r^2} - \frac{\Lambda z^2 B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{4\, r^2} \\
& - \frac{z^2 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A\left(x\right)}{8\, r^2} + \frac{z^2 e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B\left(y\right)}{8\, r^2} \\
& - \frac{\Lambda A\left(x\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F\left(r,x\right)}{8\, r^2} - \frac{\Lambda B\left(y\right) e^{\left(\Lambda^2 r^4 A(x) B(y) - \Lambda F(r,x) - \Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G\left(r,y\right)}{8\, r^2}
\end{aligned}
\tag{142}
$$

$$\Phi'_{12} = \frac{\Lambda z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) \frac{\partial}{\partial x} F(r,x)}{4\,r^2} + \frac{\Lambda z A(x)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F(r,x)}{4\,r^2}$$
$$- \frac{\Lambda z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \frac{\partial}{\partial y} G(r,y)}{4\,r^2} - \frac{\Lambda z B(y)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G(r,y)}{4\,r^2}$$
$$- \frac{z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A(x)}{8\,r^2} + \frac{z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B(y)}{8\,r^2} \tag{143}$$

$$\Phi'_{20} = -\frac{1}{8}\,\Lambda^2 r^2 B(y)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x)^2 + \frac{1}{4} i\,\Lambda^2 r^2 \sqrt{A(x)}\sqrt{B(y)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) \frac{\partial}{\partial y} B(y)$$
$$+ \frac{1}{8}\,\Lambda^2 r^2 A(x)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y)^2 + \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) \frac{\partial}{\partial x} F(r,x)}{4\,r^2}$$
$$- \frac{\Lambda^2 A(x)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x)^2}{8\,r^2} + \frac{\Lambda z^2 A(x)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F(r,x)}{4\,r^2}$$
$$- \frac{\Lambda z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \frac{\partial}{\partial y} G(r,y)}{4\,r^2} - \frac{i\,\Lambda^2 \sqrt{A(x)}\sqrt{B(y)} e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} F(r,x) \frac{\partial}{\partial y} G(r,y)}{4\,r^2}$$
$$+ \frac{\Lambda^2 B(y)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} G(r,y)^2}{8\,r^2} - \frac{\Lambda z^2 B(y)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G(r,y)}{4\,r^2}$$
$$- \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A(x)}{8\,r^2} + \frac{z^2 e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B(y)}{8\,r^2} + \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) \frac{\partial}{\partial x} F(r,x)}{4\,r^2}$$
$$- \frac{\Lambda e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \frac{\partial}{\partial y} G(r,y)}{4\,r^2} - \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A(x)}{8\,r^2} + \frac{e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B(y)}{8\,r^2} \tag{144}$$

$$\Phi'_{21} = \frac{\Lambda z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial x} A(x) \frac{\partial}{\partial x} F(r,x)}{4\,r^2} + \frac{\Lambda z A(x)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} F(r,x)}{4\,r^2}$$
$$- \frac{\Lambda z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial}{\partial y} B(y) \frac{\partial}{\partial y} G(r,y)}{4\,r^2} - \frac{\Lambda z B(y)\, e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} G(r,y)}{4\,r^2}$$
$$- \frac{z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial x)^2} A(x)}{8\,r^2} + \frac{z e^{\left(\Lambda^2 r^4 A(x)B(y)-\Lambda F(r,x)-\Lambda G(r,y)\right)} \frac{\partial^2}{(\partial y)^2} B(y)}{8\,r^2} \tag{145}$$

$$\Phi'_{22} = \Phi_{22} \tag{146}$$

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